Food Rescue Connect: Smart Food Donation and Redistribution Management System

Industry Context

Agile sprint planning & user story workshop

A CASE STUDY REPORT

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Introduction

Food waste is one of the most critical global challenges, with millions of tons of edible food being discarded every year while millions of people continue to suffer from hunger and food insecurity. Restaurants, supermarkets, hotels, catering services, and event organizers often generate surplus food due to overproduction, changing customer demand, and strict quality standards. Although much of this food remains safe for consumption, it is frequently disposed of because of the absence of an efficient system to collect and redistribute it. This not only results in economic losses but also contributes to environmental problems such as increased landfill waste and greenhouse gas emissions.

Recent advancements in technologies such as Artificial Intelligence (AI), Cloud Computing, Internet of Things (IoT), GPS, and Mobile Applications have created new opportunities to improve food donation and redistribution processes. These technologies enable real-time communication between donors, volunteers, non-governmental organizations (NGOs), and beneficiaries while supporting intelligent decision-making for food allocation, route optimization, and delivery scheduling.

FoodRescue Connect is a smart food donation and redistribution management system designed to address these challenges by providing a centralized digital platform that connects food donors, NGOs, volunteers, and beneficiaries. The system allows donors to register surplus food, monitor food quantity and expiry time, and request pickups. AI algorithms analyze donation requests, prioritize urgent food items, and determine the most efficient transportation routes to minimize delivery time and reduce food spoilage. Real-time notifications, GPS tracking, and cloud-based data management ensure transparency and efficient coordination throughout the donation process.

The proposed system aims to significantly reduce food waste, improve food accessibility for vulnerable communities, enhance operational efficiency for charitable organizations, and promote sustainable food recovery practices. By integrating modern digital technologies with social welfare initiatives, FoodRescue Connect contributes to environmental sustainability, community development, and the achievement of Sustainable Development Goals (SDGs), particularly Zero Hunger (SDG 2) and Responsible Consumption and Production (SDG 12). This architecture design report presents the system requirements, software architecture, component interactions, quality attributes, and architectural decisions for developing a scalable, reliable, and secure smart food donation management platform.

The implementation of FoodRescue Connect not only addresses food insecurity but also contributes to environmental sustainability by reducing landfill waste and greenhouse gas emissions caused by discarded food. Furthermore, it supports the achievement of sustainable development goals related to zero hunger, responsible consumption and production, and sustainable communities. By integrating modern technologies into food supply chain management, the proposed system offers a practical, scalable, and efficient solution for creating a smarter and more sustainable food donation ecosystem.

Furthermore, the success of a smart food donation system depends not only on advanced technologies but also on active participation and collaboration among all stakeholders. Food donors, charitable organizations, volunteers, government agencies, and local communities must work together to ensure that surplus food is collected and distributed efficiently. A centralized digital platform enhances transparency, accountability, and trust by providing real-time updates on donation status, food quality, pickup schedules, and delivery progress.

Problem Statement

Food waste is a major global issue, with large quantities of safe and nutritious food being discarded every day by restaurants, supermarkets, hotels, event organizers, and food manufacturers. At the same time, many people continue to face hunger and food insecurity due to the unequal distribution of available food resources. The primary reason for this problem is the lack of an efficient and coordinated system that connects food donors with charitable organizations and beneficiaries.

Most existing food donation systems rely on manual communication methods such as phone calls, emails, or messaging applications, which often lead to delays in food collection and distribution. As a result, donated food may spoil before it reaches those in need, reducing the effectiveness of food recovery efforts. In addition, NGOs and volunteers face difficulties in identifying nearby food donations, assigning pickup tasks, planning efficient transportation routes, and tracking deliveries in real time.

Current platforms also make limited use of modern technologies such as Artificial Intelligence (AI), GPS, Cloud Computing, and real-time analytics. Without intelligent route optimization, food priority prediction, and automated notifications, transportation costs increase, response times become longer, and available food is not distributed fairly among beneficiaries.

Therefore, there is a need for an intelligent and centralized **FoodRescue Connect** system that integrates food donors, NGOs, volunteers, and beneficiaries into a single digital platform. The system should use AI to predict food availability, prioritize urgent donations, optimize transportation routes, monitor food quality and expiry time, and provide real-time tracking and notifications

Another major challenge is the absence of intelligent decision-making in existing food donation systems. Most platforms do not utilize Artificial Intelligence (AI) to predict food availability, estimate food shelf life, prioritize urgent donations, or optimize transportation routes. Without AI-based planning, food collection and delivery become inefficient, increasing transportation costs, travel time, and fuel consumption. Poor route planning can also lead to delayed deliveries, causing food to spoil before reaching the beneficiaries.

Food quality and safety are additional concerns. Existing systems generally provide limited facilities to monitor food freshness, storage conditions, quantity, and expiry time. Without proper monitoring, there is a higher risk of distributing food that is no longer suitable for consumption, which may lead to health and safety issues. Furthermore, organizations often lack real-time visibility into the status of donations, making it difficult to track whether food has been collected, delivered, or distributed successfully.

Coordination among stakeholders is another significant issue. Food donors, NGOs, volunteers, beneficiaries, and government agencies often work independently with limited communication and data sharing. This fragmented approach leads to duplicate donation requests, unequal food distribution, inefficient resource utilization, and poor transparency throughout the donation process. In addition, administrators face difficulties in maintaining records, generating reports, measuring food waste reduction, and evaluating the overall effectiveness of food recovery initiatives.

The lack of analytical tools also limits long-term planning. Existing systems provide minimal reporting and data analysis, making it difficult for organizations to understand donation trends, identify high-demand areas, evaluate volunteer performance, and measure environmental impact.

Scope of the System

The system provides services to multiple users including donors, NGOs, volunteers, and beneficiaries.

The scope includes:

- Food donation registration
- Donation request management
- Volunteer assignment
- Pickup scheduling
- GPS route optimization
- Delivery tracking
- Food quality monitoring
- Real-time notifications
- Analytics dashboard
- Report generation

Future enhancements may include IoT sensors, blockchain verification, and machine learning-based food demand prediction.

Existing System

The current food donation process is mostly manual.

Features

- Phone calls
- WhatsApp communication
- Manual volunteer assignment
- Paper-based records

Disadvantages

- Slow communication
- Food spoilage
- Duplicate requests
- Poor transparency
- No tracking system
- Difficult report generation

6. Proposed System

FoodRescue Connect introduces a centralized cloud platform where all stakeholders work together digitally.

The system automatically:

- Registers food donors
- Accepts donation requests
- Calculates food priority
- Predicts expiry
- Assigns nearest volunteer
- Optimizes delivery route
- Tracks delivery
- Updates beneficiaries

- Generates reports

Artificial Intelligence helps distribute food efficiently while minimizing waste.

7. Functional Requirements

The system shall allow:

User Management

- User Registration
- Login
- Authentication
- Profile Management

Donor Module

- Add food donation
- Upload food details
- Mention expiry time
- Upload food images

NGO Module

- View nearby donations
- Accept donation
- Manage beneficiaries

Volunteer Module

- Receive pickup request
- Accept delivery
- Update delivery status

Beneficiary Module

- View available food
- Receive food
- Confirm receipt

Notification Module

- SMS alerts
- Email alerts
- Push notifications

AI Module

- Route optimization
- Food priority prediction
- Volunteer assignment

Reporting Module

- Donation reports
- Waste reduction reports
- Monthly analytics

8. Non-Functional Requirements

The system should provide:

Performance

Fast response within 2–3 seconds.

Scalability

Support thousands of simultaneous users.

Reliability

Operate continuously without failure.

Availability

24×7 cloud availability.

Security

- User authentication
- Data encryption
- Secure APIs

Maintainability

Easy software updates.

Usability

Simple user interface.

Portability

Accessible through web and mobile applications.

9. Stakeholders

Food Donors

Restaurants, Hotels, Supermarkets

NGOs

Food distribution organizations

Volunteers

Pickup and delivery staff

Beneficiaries

People receiving food

Government

Food safety monitoring

Administrator

System management

10. System Architecture

The proposed system follows a **Hybrid Microservices Architecture**.

Main Layers

Presentation Layer

- Mobile App
- Web Portal

API Gateway

- Request validation
- Authentication

Service Layer

- User Service
- Donation Service
- Volunteer Service
- Notification Service
- AI Service
- Reporting Service

Database Layer

- MySQL
- MongoDB

External Services

- Google Maps API
- SMS Gateway
- Email Server
- Cloud Storage

11. Major Components

User Management

Handles registration, login, and authentication.

Donation Management

Stores food information including quantity, expiry date, and pickup location.

Volunteer Management

Assigns volunteers automatically based on location.

AI Engine

Predicts:

- Best volunteer
- Shortest route
- Food priority

Notification System

Sends

- Pickup alerts
- Delivery alerts
- Expiry alerts

Tracking System

Tracks food movement from donor to beneficiary.

Reporting Module

Generates

- Daily reports
- Monthly reports
- Donation statistics

12. System Workflow

Step 1

Donor logs into the system.

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Step 2

Adds food donation details.

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Step 3

AI checks expiry time and priority.

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Step 4

Nearby NGOs receive notification.

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Step 5

Nearest volunteer is assigned.

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Step 6

Volunteer collects food.

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Step 7

GPS tracks delivery.

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Step 8

Beneficiary receives food.

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Step 9

System updates records.

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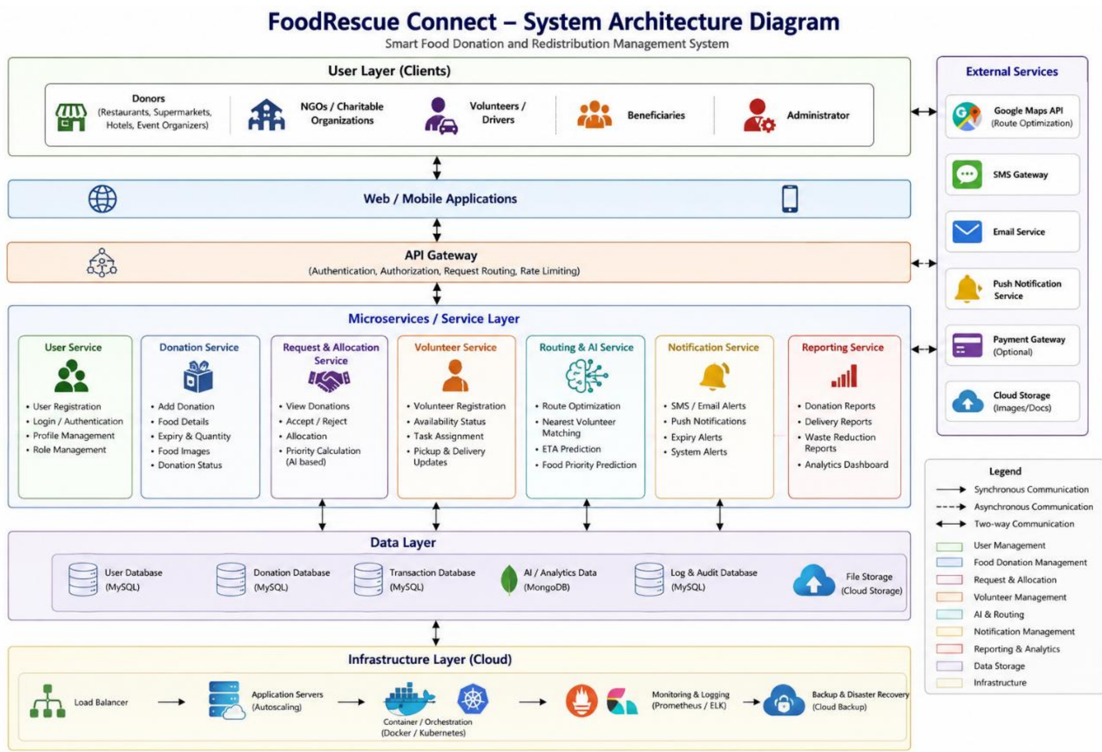
Step 10

Reports are generated.



Architecture Diagram Explanation

The FoodRescue Connect system uses a Hybrid Microservices Architecture, where different modules work together to manage food donation and redistribution. Each module has a specific responsibility, making the system easy to manage, secure, and scalable.

1. Users

The users of the system include Donors, NGOs, Volunteers, Beneficiaries, and Administrators. They access the system through a web or mobile application to perform their respective tasks such as donating food, accepting requests, delivering food, or managing the platform.

2. Web/Mobile Application

This is the interface through which users interact with the system. It allows users to register, log in, donate food, receive notifications, track deliveries, and view reports.

3. API Gateway

The API Gateway acts as the entry point for all user requests. It verifies users, checks permissions, and forwards requests to the appropriate service securely.

4. Service Layer

The Service Layer contains different modules that perform the main functions of the system.

- User Service: Manages user registration, login, and profiles.
- Donation Service: Records food donation details such as quantity, expiry time, and location.
- Volunteer Service: Assigns volunteers for food collection and delivery.
- AI & Route Optimization Service: Uses Artificial Intelligence to select the nearest volunteer and find the shortest delivery route.
- Notification Service: Sends alerts through SMS, email, or mobile notifications.
- Reporting Service: Generates reports on donations, deliveries, and food waste reduction.

5. Database Layer

The database stores all important information, including user details, donation records, volunteer information, delivery status, and system reports. Cloud storage is used to save images and backup data securely.

6. External Services

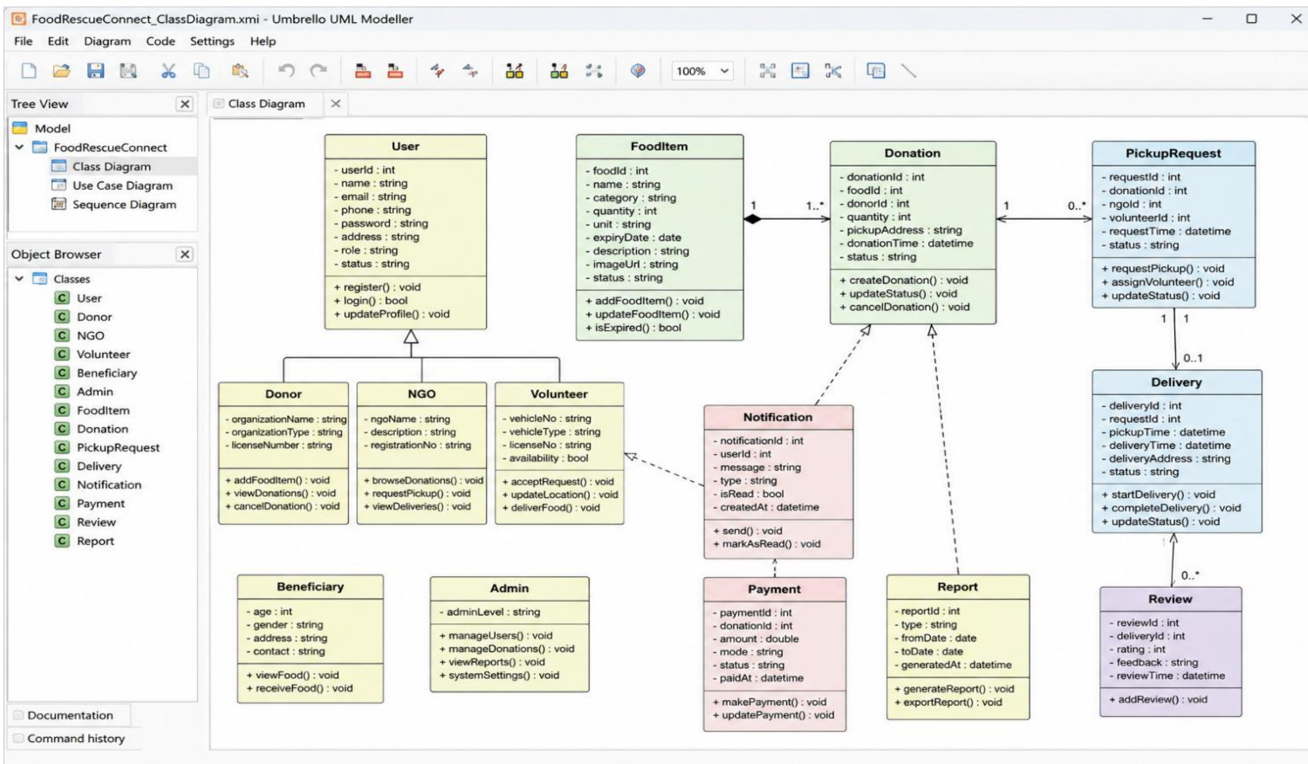
The system connects with external services such as Google Maps for GPS navigation and route optimization, SMS and Email Services for notifications, and Cloud Storage for storing files and reports.

7. System Workflow

The process begins when a donor adds surplus food through the application. The request is sent to the API Gateway, which forwards it to the Donation Service. The AI Service analyzes the food details and identifies the nearest NGO and volunteer.

The Notification Service informs the volunteer about the pickup request. The volunteer collects the food and delivers it to the beneficiary using the optimized route provided by Google Maps. Finally, the system updates the delivery status and generates reports for the administrator.

Class Diagram

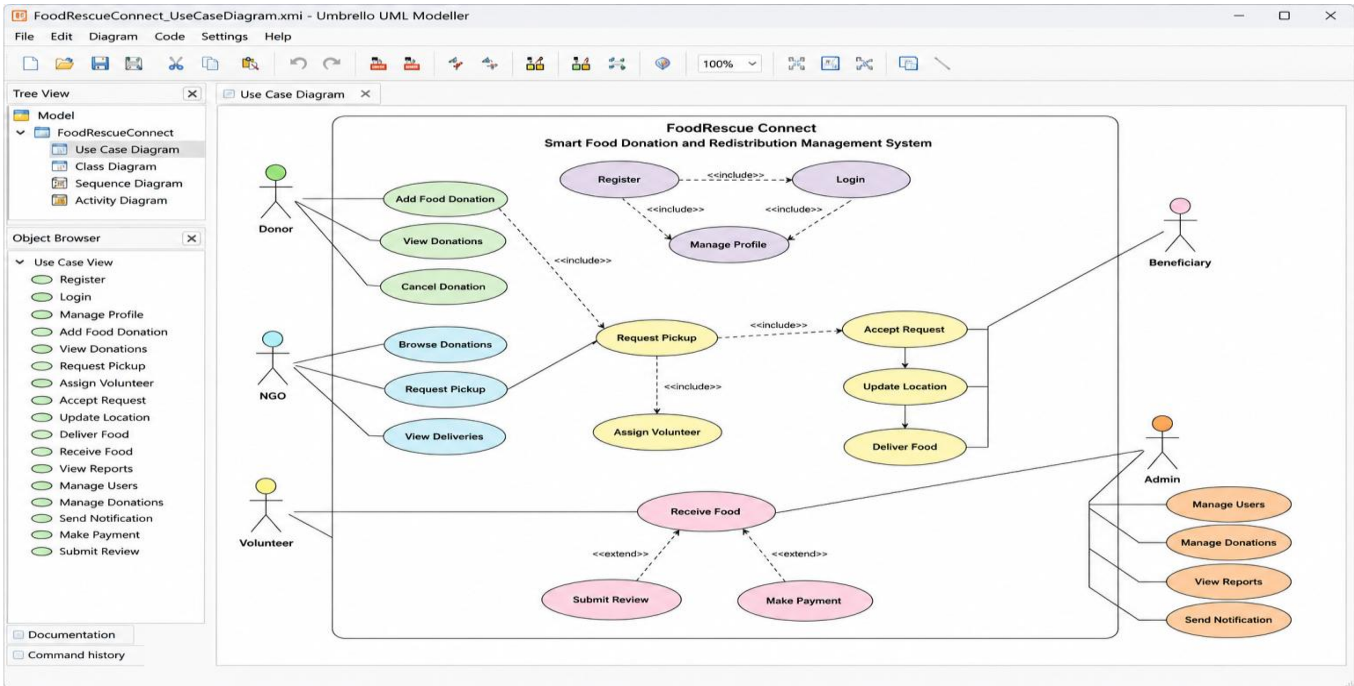


The class diagram of the FoodRescue Connect: Smart Food Donation and Redistribution Management System illustrates the static structure of the system by showing the main classes, their attributes, methods, and relationships. The User class serves as the parent class and contains common information such as user ID, name, email, phone number, password, address, role, and status. It provides basic functions like registration, login, and profile management. The Donor, NGO, and Volunteer classes inherit from the User class, allowing them to share common user features while performing their specific responsibilities. Donors can add food items, view donations, and cancel donations. NGOs can browse available food donations, request pickups, and monitor deliveries. Volunteers are responsible for accepting pickup requests, updating their location, and delivering food to beneficiaries. The Beneficiary class represents the people who receive the donated food, while the Admin class manages users, donations, reports, and system settings.

The FoodItem class stores detailed information about donated food, including its name, category, quantity, expiry date, description, and status. Every donated food item is associated with the Donation class, which records donation details such as donor information, pickup address, donation time, quantity, and donation status. Once a donation is created, the PickupRequest class generates a request for food collection and assigns an available volunteer. The Delivery class tracks the transportation process by recording pickup time, delivery time, destination address, and delivery status. Throughout the process, the Notification class sends important updates such as pickup confirmations, delivery alerts, and system notifications through SMS, email, or mobile notifications.

The system also includes supporting classes such as Payment, Report, and Review. The Payment class manages optional payment transactions and stores payment details if required. The Report class generates reports related to food donations, deliveries, and food waste reduction, helping administrators analyze system performance. The Review class allows beneficiaries to provide ratings and feedback after receiving food, improving service quality. The relationships among these classes represent the complete workflow of the system, where donors register food donations, NGOs request the food, volunteers collect and deliver it, beneficiaries receive the food, and administrators monitor the entire process. This class diagram provides a clear and organized representation of the system, making it easier to understand, develop, maintain, and expand in the future.

Use CASE Diagram



- The **Use Case Diagram** shows how different users interact with the **FoodRescue Connect** system.
- The main actors are **Donor, NGO, Volunteer, Beneficiary, and Admin**.
- The **Donor** can register, log in, add food donations, view donations, and cancel donations.
- The **NGO** can browse available donations, request food pickup, and view delivery details.
- The **Volunteer** accepts pickup requests, updates location, and delivers food to beneficiaries.
- The **Beneficiary** receives the donated food and can submit feedback or reviews after delivery.
- The **Admin** manages users, donations, reports, and sends notifications to system users.
- The diagram also shows relationships such as **«include»** and **«extend»**, representing dependent and optional use cases.
- Overall, the use case diagram explains the complete workflow of the FoodRescue Connect system, from food donation to successful delivery and administration.

13. Quality Attributes

Scalability

Supports growing numbers of users using cloud infrastructure.

Reliability

Automatic backup and recovery ensure uninterrupted operation.

Performance

AI algorithms reduce delivery time.

Security

- Role-based access
- HTTPS communication
- Encrypted passwords

Maintainability

Independent services simplify updates.

Availability

Cloud deployment ensures 24×7 service.

14. Advantages

- Reduces food waste
- Faster food delivery
- Better communication
- AI-based optimization
- GPS tracking

- Real-time notifications
- Digital reports
- Better transparency
- Environmental sustainability
- Improved food accessibility

15. Limitations

- Requires internet connection
- GPS accuracy may vary
- Initial setup cost
- Depends on donor participation
- AI prediction accuracy depends on available data

16. Future Enhancements

Future versions can include:

- IoT temperature sensors
- Blockchain for food traceability
- Voice assistant support
- AI demand forecasting
- Drone-based food delivery
- QR code verification
- Multi-language support
- Integration with government food programs

17. Expected Outcomes

The proposed system is expected to:

- Reduce edible food waste.
- Improve food availability for needy people.

- Enable faster collection and delivery.
- Optimize transportation using AI.
- Improve coordination among donors, NGOs, and volunteers.
- Provide transparent real-time tracking.
- Support environmental sustainability.
- Build a scalable smart food donation ecosystem.

18. Conclusion

FoodRescue Connect is an intelligent food donation and redistribution platform that helps reduce food waste while improving food security. By integrating AI, GPS, cloud computing, and real-time notifications, the system ensures surplus food is collected and delivered quickly and efficiently. The proposed architecture provides scalability, reliability, security, and maintainability, making it suitable for large-scale deployment. This solution supports sustainable development goals by minimizing waste, reducing environmental impact, and improving community welfare.

This content aligns with the architecture design report structure required in your assignment, including system requirements, architecture selection, component interaction, quality attributes, and architectural justification.